

Understanding Analysis, 2nd Edition

Chapter 6 - Section 6.2 - Uniform Convergence of a Sequence of Functions

1. Let

$$f_n(x) = \frac{nx}{1 + nx^2}$$

- (a) Find the pointwise limit of (f_n) for all $x \in (0, \infty)$.
- (b) Is the convergence uniform on $(0, \infty)$?
- (c) Is the convergence uniform on $(0, 1)$?
- (d) Is the convergence uniform on $(1, \infty)$?

Solution:

- (a) Intuitively, as n grows the $+1$ term starts to not matter, so it should converge to $\frac{nx}{nx^2} = \frac{1}{x}$. Formally

$$\lim_{n \rightarrow \infty} \frac{nx}{1 + nx^2} = \lim_{n \rightarrow \infty} \frac{x}{1/n + x^2} = 1/x$$

- (b) No. Given $\epsilon > 0$, let N such that for all $n \geq N$, $|f_n(x) - \frac{1}{x}| < \epsilon$. Or in other words

$$\left| \frac{nx}{1 + nx^2} - \frac{1}{x} \right| = \frac{1}{x(1 + nx^2)} < \epsilon$$

For this to be uniformly convergent, the N needs to work for all x . But the lhs is unbounded as $x \rightarrow 0$; there's always a small enough x such that the lhs is eventually greater than ϵ .

- (c) No, for the same reason as part (b). We can always choose small enough x to break the uniform convergence.
- (d) Now the answer is yes. Reusing the result from part (b), for $x > 1$, the value is upper-bounded by $x = 1$

$$\frac{1}{x(1 + nx^2)} < \frac{1}{1 + n}$$

This doesn't depend on x , so we can simply choose $N \geq \frac{1}{\epsilon} - 1$.

2. (a) Define a sequence of functions on $\mathbf{R}$ by

$$f_n(x) = \begin{cases} 1 & \text{if } x = 1, \frac{1}{2}, \frac{1}{3}, \dots, \frac{1}{n} \\ 0 & \text{otherwise} \end{cases}$$

and let f be the pointwise limit of f_n .

Is each f_n continuous at zero? Does $f_n \rightarrow f$ uniformly on $\mathbf{R}$? Is f continuous at zero?

- (b) Repeat this exercise using the sequence of functions

$$g_n(x) = \begin{cases} x & \text{if } x = 1, \frac{1}{2}, \frac{1}{3}, \dots, \frac{1}{n} \\ 0 & \text{otherwise} \end{cases}$$

- (c) Repeat once more with

$$h_n(x) = \begin{cases} 1 & \text{if } x = \frac{1}{n} \\ x & \text{if } x = 1, \frac{1}{2}, \frac{1}{3}, \dots, \frac{1}{n-1} \\ 0 & \text{otherwise} \end{cases}$$

In each case, explain how the results are consistent with the Continuous Limit Theorem (Theorem 6.2.6).

Solution:

- (a) Yes, each f_n is continuous at zero. Simply choose $\delta < \frac{1}{n}$, then $f(x) = 0$ for $x < \delta$.

To answer uniform convergence, let's first define f . For x not of the form $\frac{1}{n}$, $f(x)$ is clearly 0. For x of the form $\frac{1}{n}$, then for all $m \geq n$, $f_m(x) = 1$ i.e. eventually all $f_n(x)$ becomes 1. So f is

$$f(x) = \begin{cases} 1 & \text{if } x = \frac{1}{n} \text{ for } n \in \mathbf{N} \\ 0 & \text{otherwise} \end{cases}$$

It's not uniformly convergent. Let $\epsilon < 1$. Now say we choose $N = 5$, then for $x = \frac{1}{6}$, we have $f_5(\frac{1}{6}) = 0$ but $f(\frac{1}{6}) = 1$. In general, no single N can work for all x because we can choose $x = \frac{1}{N+1}$, then $|f_N(x) - f(x)| = 1 > \epsilon$.

Finally, f is not continuous at zero. We have $f(0) = 0$. But since $\frac{1}{n} \rightarrow 0$, there's always x of the form $\frac{1}{n}$ close enough to zero such that $f(x) = 1$.

- (b) Each g_n is continuous at zero. Same argument as part (a), choose $\delta < \frac{1}{n}$ then $|x| < \delta \implies g_n(x) = 0$.

Using a very similar argument as part (a), the limit function g is

$$g(x) = \begin{cases} x & \text{if } x = \frac{1}{n} \text{ for } n \in \mathbf{N} \\ 0 & \text{otherwise} \end{cases}$$

But unlike part(a), this is uniformly convergent. The idea is to choose N big enough such that even if $g_N(x) = 0$, then x needs to be really small, hence $|g_N(x) - g(x)| = |0 - x| = |x|$ is really small as well.

Formally, choose $N > \frac{1}{\epsilon}$. Then for $n \geq N$, either $g_n(x) = g(x)$ or if $g_n(x) = 0$, then $x = \frac{1}{m}$ with $m > n$. But since $n > \frac{1}{\epsilon}$, if $m > n$ then $\frac{1}{m} < \epsilon$, hence $|g_n(\frac{1}{m}) - g(\frac{1}{m})| = \frac{1}{m} < \epsilon$.

g is continuous at zero, as $\frac{1}{n}$ converges to zero.

- (c) Each h_n is continuous at zero. Choose $\delta < \frac{1}{n}$, then $|x| < \delta \implies h_n(x) = 0$.

Using a similar argument as part (a), for each x of the form $\frac{1}{n}$, we have $h_m(x) = x = \frac{1}{n}$ for all $m > n$. So h is

$$h(x) = \begin{cases} x & \text{if } x = \frac{1}{n} \text{ for } n \in \mathbf{N} \\ 0 & \text{otherwise} \end{cases}$$

h is the same as g from part(b), but in this case it's not uniformly convergent. The issue here is that

$$h_N(\frac{1}{N}) - h(\frac{1}{N}) = 1 - \frac{1}{N}$$

So as $x = \frac{1}{n}$ gets smaller, we need to choose increasingly larger N .

More formally, let $\epsilon < \frac{1}{2}$ and $x = \frac{1}{2}$. Choose $N \geq 3$. Then for $n \geq N$, we have

$$h_n(\frac{1}{2}) - h(\frac{1}{2}) = \frac{1}{2} - \frac{1}{2} = 0$$

But this same N cannot for $x = \frac{1}{N+1}$ because for $n = N+1$ we have

$$h_{N+1}(\frac{1}{N+1}) - h(\frac{1}{N+1}) = 1 - \frac{1}{N+1}$$

And since $N \geq 3$, then $\frac{1}{N+1} \leq \frac{1}{4}$, hence

$$1 - \frac{1}{N+1} \geq 1 - \frac{1}{4} > \epsilon$$

Finally, h is continuous at zero as it's the same as g in part(b).

3. TODO
4. TODO. It depends on Exercise 5.2.8 which I haven't done yet.
5. Using the Cauchy Criterion for convergent sequences of real numbers (Theorem 2.6.4), supply a proof for Theorem 6.2.5. (First, define a candidate for $f(x)$, and then argue that $f_n \rightarrow f$ uniformly).

Solution: The proof strategy is very similar to Theorem 2.6.4.

Necessity: uniform convergence $\implies$ Cauchy

There exists N such that for all $n \geq N$, $|f_n(x) - f(x)| < \epsilon/2$ for all x . Then for any $m, n \geq N$ and any x we have

$$|f_n(x) - f_m(x)| \leq |f_n(x) - f(x)| + |f(x) - f_m(x)| < \epsilon/2 + \epsilon/2 = \epsilon$$

Satisfying Cauchy criterion.

Sufficiency: Cauchy $\implies$ uniform convergence

The Cauchy criterion doesn't say what the limit function is, so we need to define it first. For a **particular** x , the sequence of real numbers $f_1(x), f_2(x), \dots$ is a Cauchy sequence, so it converges to a limit L_x . Define f to be

$$f(x) = L_x$$

Now we prove uniform convergence. By the Cauchy hypothesis (for sequence of functions), there exists N such that for all $n, m \geq N$, we have

$$|f_n(x) - f_m(x)| < \epsilon/2$$

for any x .

Also, by the fact that $f_1(x), f_2(x), \dots \rightarrow L_x$, there exists an $k \geq N$ such that $|f_k(x) - L_x| < \epsilon/2$. The value of k depends on x , but that's okay because for any x , $k \geq N$ and N does not depend on x .

Combining these two inequalities, we satisfy the condition for uniform convergence. For any $n \geq N$, we have

$$|f_n(x) - L_x| \leq |f_n(x) - f_k(x)| + |f_k(x) - L_x| < \epsilon/2 + \epsilon/2 = \epsilon$$

for any x .